



2.12

Logarithmic Function Manipulation

Core-to-Challenge Worksheet and AP Practice

AP Precalculus - Unit 2 | ECHS Mathematics | Teacher: Mohammad Abu-Ghuwaleh

Student Information

Name: _____ Class: _____ Date: _____

Directions

Show mathematical evidence. Retain domain restrictions, label graphs, include units, and write complete interpretations. Calculator use is permitted only when the item explicitly requires approximation, regression, or numerical solving.

Readiness

1. Write one precise sentence explaining this principle: Product property requires positive factors in the real logarithm setting.

2. Write one precise sentence explaining this principle: Power property brings an exponent down as a multiplier.

3. Write one precise sentence explaining this principle: Change of base allows evaluation using common or natural logs.

4. Write one precise sentence explaining this principle: Algebraic simplification must preserve domain restrictions.

Core

5. Expand $\ln \frac{x^3\sqrt{x-1}}{(x+2)^2}$.
6. Condense $2 \log_5 x - \frac{1}{2} \log_5 y$.
7. Evaluate $\log_2 7$ using natural logarithms.
8. State what the formula $\log_b(MN) = \log_b M + \log_b N$ tells you and name any required restriction.
9. State what the formula $\log_b(M/N) = \log_b M - \log_b N$ tells you and name any required restriction.
10. State what the formula $\log_b(M^p) = p \log_b M$ tells you and name any required restriction.

Medium

11. Expand, condense, or change base as appropriate: $\ln(x^3y^2)$.
12. Expand, condense, or change base as appropriate: $\log_2 \frac{a^4}{b\sqrt{c}}$.
13. Expand, condense, or change base as appropriate: $2 \ln x - 3 \ln y$.
14. Expand, condense, or change base as appropriate: $\log_5 m + \log_5 n - \log_5 p$.

15. Expand, condense, or change base as appropriate: $\log_7 12$.

Hard

16. Expand $\ln \sqrt{\frac{x^2-1}{x^4}}$ with domain conditions.

17. Condense $\frac{1}{3} \log x + 2 \log(x-1) - \log(x+4)$.

18. Explain why $\ln(x+y) \neq \ln x + \ln y$.

19. Show $\log_4 x = \frac{1}{2} \log_2 x$.

Difficult

20. Simplify $\log_b(b^{2x} \sqrt[3]{b^{x-1}})$.

21. Find domain preserved by $\ln(x^2 - 4) - \ln(x - 2)$ before condensing/simplifying.

22. Prove product property from exponential definitions.

Challenge / HOT

23. Find all x in the original domain satisfying $\ln(x^2) = 2 \ln(x - 1)$.

24. Explain why absolute values appear in identities like $\ln(x^2) = 2 \ln |x|$.

AP-Style Multiple Choice

25. $\log(MN) =$
 (A) $\log M \log N$
 (B) $\log M + \log N$
 (C) $\log(M + N)$
 (D) $M \log N$

26. $\log(M/N) =$
 (A) $\log M - \log N$
 (B) $\log M / \log N$
 (C) $\log M + \log N$
 (D) $N - M$
27. $\log(M^p) =$
 (A) $(\log M)^p$
 (B) $p \log M$
 (C) $M \log p$
 (D) $\log M + p$
28. Change of base for $\log_b x$ is
 (A) $\ln b / \ln x$
 (B) $\ln x / \ln b$
 (C) $\ln(x/b)$
 (D) $b \ln x$
29. Log properties apply to
 (A) sums directly
 (B) products/quotients/powers
 (C) negative arguments
 (D) base 1
30. When expanding $\ln x^2$ for real x not zero, use
 (A) $2 \ln x$
 (B) $2 \ln |x|$
 (C) $\ln 2x$
 (D) $x \ln 2$

AP-Style Free Response

- 31.** Given $E = \ln \left(\frac{x^4 \sqrt{y-2}}{(x+1)^3} \right)$. (a) State domain. (b) Expand. (c) Re-condense your expansion and explain equivalence restrictions.

- 32.** Prove change-of-base formula and use it to show $\log_b x \log_x b = 1$.

Reflection

Identify one item where a domain restriction, unit, assumption, or representation changed your reasoning. Explain in 2-3 sentences.

Original ECHS questions. Selected concepts are informed by Pearson and Holt Precalculus; no textbook exercises are reproduced.